Allosteric Inhibitors of Coagulation Proteases

Project Number
5R01HL090586-08

Contact PI/Project Leader
DESAI, UMESH RAMANLAL

Awardee Organization
VIRGINIA COMMONWEALTH UNIVERSITY

Description

Abstract Text

DESCRIPTION (provided by applicant): This project attempts to develop novel molecules as allosteric anticoagulants of coagulation proteases, especially thrombin and factor XIa. The hypothesis is that targeting these proteases through allosteric disruption of their catalytic triad results in controlled inhibition, which is likely to not suffer from bleeding consequences. All current anticoagulants including the heparins, warfarin, dabigatran and rivaroxaban suffer from enhanced risk of bleeding. The past grant period resulted in two groups of highly promising inhibitors of thrombin and factor XIa. The thrombin inhibitors displayed regulatory features (maximal inhibition in the region of 50 - 75%), which is likely to maintain hemostasis, especially under prophylactic application. Likewise, factor XIa inhibitors are likely to be devoid of bleeding because of the inherent mechanism of action of factor XIa, which is its action on the thrombotic process while leaving hemostatic process intact. In addition, the allosteric inhibitors are predicted to possess a higher specificity of action, which would reduce off target effects. Thus, the two groups of inhibitors are predicted to lead to a safer anticoagulant therapy. In fact, we have found that one of our designed molecules showed marked anticoagulant activities in FeCl3-induced thrombosis in the carotid artery mouse model without affecting bleeding time supporting the hypothesis. In this grant period, we intend to I) design more potent, selective regulators of human thrombin; II) develop more potent, selective inhibitors of human factor XIa and III) perform pre-clinical studies on promising advanced thrombin regulators and FXIa inhibitors.

Public Health Relevance Statement

PUBLIC HEALTH RELEVANCE: Thrombosis is a leading cause of morbidity and mortality associated with arterial and venous disorders. Nearly 1 in 3 individuals in the US suffer from thrombotic disorders. The current anticoagulants including warfarin, heparin, dabigatran and rivaroxaban suffer from significant risk of bleeding and other agent-specific problems including unpredictable pharmacokinetics and drug-drug interactions. The goal of this project is to develop safer anticoagulant drug based on our initials results with allosteric inhibition of thrombin and factor XIa.

NIH Spending Category

Biotechnology | Cardiovascular | Hematology

Project Terms

Adult | Affect | Anticoagulant therapy | Anticoagulants | Anticoagulation | Antidotes | Antithrombin III | Benzofurans | Binding | Binding Sites | Bioavailable | Biological Availability | Bleeding time procedure | Carotid Arteries | Coagulation Process | Disease | Drug Delivery Systems | Drug Design | Drug Interactions | Drug Kinetics | Electrostatics | Elements | Equilibrium | Factor XIa | Fibrinolytic Agents | Functional disorder | Generations | Glycosaminoglycans | Goals | Grant | Hemorrhage | Hemostatic Agents | Hemostatic function | Heparin | Heparitin Sulfate | Human | In Vitro | Individual | Investigational Drugs | Lead | Legal patent | Libraries | Mammals | Modeling | Morbidity - disease rate | Oral | Paper | Pathway interactions | Peptide Hydrolases | Pharmaceutical Preparations | Physicians | Primates | Process

Read Less

Property Regulation Research Research Personnel Risk Safety Specificity Surgeon System Tail Therapeutic Thrombin Thromboplastin Thrombosis Time Toxic effect Triad Acrylic Resin Unspecified or Sulfate Ion Sulfates Venous Warfarin Work X-Ray Crystallography analog base carboxylate clinically relevant design dimer high throughput screening in vivo inhibitor/antagonist innovation lead sulfate mimetics mortality mouse model novel pre-clinical preclinical study programs prophylactic public health relevance scaffold structural biology

Details

Contact PI/ Project Leader

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DESAI, UMESH RAMANLAL
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Not Applicable

Program Official

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VIRGINIA COMMONWEALTH UNIVERSITY
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RICHMOND
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UNITED STATES (US)
Department Type
PHARMACOLOGY
Organization Type
SCHOOLS OF PHARMACY
State Code
VA
Congressional District
04

Other Information

Opportunity Number
PA-13-302
Study Section
Hemostasis and Thrombosis Study Section. Committee was terminated on 11/30/2020. [HT]
Fiscal Year
2018
Award Notice Date
09-August-2018
Administering Institutes or Centers
National Heart Lung and Blood Institute
CFDA Code
839
DUNS Number
105300446
UEI
MLQFL4JSSAA9
Project Start Date **01-October-2007**
Project End Date **31-July-2020**
Budget Start Date **01-August-2018**
Budget End Date **31-July-2020**

Project Funding Information for 2018

Total Funding
$512,380
Direct Costs
$352,380
Indirect Costs
$160,000

Year	Funding IC	FY Total Cost by IC
2018	National Heart Lung and Blood Institute	$512,380

NIH Categorical Spending

Click here for more information on NIH Categorical Spending

Funding IC	FY Total Cost by IC	NIH Spending Category
NATIONAL HEART, LUNG, AND BLOOD INSTITUTE	$512,380	Biotechnology; Cardiovascular; Hematology

Sub Projects

No Sub Projects information available for 5R01HL090586-08

Publications

Export › Disclaimer

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iC
Preferential recognition and antagonism of SARS-CoV-2 spike glycoprotein binding to 3- O -sulfated heparan sulfate.					
bioRxiv : the preprint server for biology 2020 Oct 08;	Tiwari, Vaibhav; Tandon, Ritesh; Sankaranarayanan, Nehru Viji; Beer, Jacob C; Kohlmeir, Ellen K; Swanson-Mungerson, Michelle; Desai, Umesh R	2020			iCite 1.85
Sulfated Non-Saccharide Glycosaminoglycan Mimetics as Novel Drug Discovery Platform for Various Pathologies.					
Current medicinal chemistry 2020; 27 (21) 3412-3447	Afosah, Daniel K; Al-Horani, Rami A	2020			iCite 1.24
Inhibition of Human Cytomegalovirus Entry into Host Cells Through a Pleiotropic Small Molecule.					
International journal of molecular sciences 2020 Feb 29; 21 (5)	Elste, James; Kaltenbach, Dominik; Patel, Vraj R; Nguyen, Max T; Sharthiya, Harsh; Tandon, Ritesh; Mehta, Satish K; Volin, Michael V; Fornaro, Michele; Tiwari, Vaibhav; Desai, Umesh R View Less	2020			iCite 0.65
Discovering small-molecule therapeutics against SARS-CoV-2.					
Drug discovery today 2020 08; 25 (8) 1535-1544	Tiwari, Vaibhav; Beer, Jacob C; Sankaranarayanan, Nehru Viji; Swanson-Mungerson, Michelle; Desai, Umesh R	2020			iCite 5.24
Studies on fragment-based design of allosteric inhibitors of human factor XIa.					
Bioorganic & medicinal chemistry 2020 12 01; 28 (23) 115762	Boothello, Rio S; Sankaranarayanan, Nehru Viji; Afosah, Daniel K; Karuturi, Rajesh; Al-Horani, Rami A; Desai, Umesh R	2020			iCite 0.56
Discovery of Sulfated Small Molecule Inhibitors of Matrix Metalloproteinase-8.					
Biomolecules 2020 08 09; 10 (8)	Morla, Shravan; Desai, Umesh R	2020			iCite 0.70
Rigorous analysis of free solution glycosaminoglycan dynamics using simple, new tools.					
Glycobiology 2020 07 16; 30 (8) 516-527	Nagarajan, Balaji; Sankaranarayanan, Nehru Viji; Desai, Umesh R	2020			iCite 1.10
A Unique Nonsaccharide Mimetic of Heparin Hexasaccharide Inhibits Colon Cancer Stem Cells via p38 MAP Kinase Activation.					
Molecular cancer therapeutics 2019 Jan; 18 (1) 51-61	Boothello, Rio S; Patel, Nirmita J; Sharon, Chetna; Abdelfadiel, Elsamani I; Morla, Shravan; Brophy, Donald F; Lippman, H Robert; Desai, Umesh R; Patel, Bhaumik B View Less	2019			iCite 1.86
On the Process of Discovering Leads That Target the Heparin-Binding Site of Neutrophil Elastase in the Sputum of Cystic Fibrosis Patients.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Journal of medicinal chemistry 2019 06 13; 62 (11) 5501-5511	Morla, Shravan; Sankaranarayanan, Nehru Viji; Afosah, Daniel K; Kumar, Megh; Kummarapurugu, Apparao B; Voynow, Judith A; Desai, Umesh R	2019			1.07
Perspective on computational simulations of glycosaminoglycans.					
Wiley interdisciplinary reviews. Computational molecular science 2019 Mar-Apr; 9 (2)	Nagarajan, Balaji; Sankaranarayanan, Nehru Viji; Desai, Umesh R	2019			1.44
A synthetic heparin mimetic that allosterically inhibits factor XIa and reduces thrombosis in vivo without enhanced risk of bleeding.					
Journal of thrombosis and haemostasis : JTH 2019 Dec; 17 (12) 2110-2122	Al-Horani, Rami A; Abdelfadiel, Elsamani I; Afosah, Daniel K; Morla, Shravan; Sistla, Jyothi C; Mohammed, Bassem; Martin, Erika J; Sakagami, Masahiro; Brophy, Donald F; Desai, Umesh R View Less	2019			1.64
A synthetic glycosaminoglycan mimetic blocks HSV-1 infection in human iris stromal cells.					
Antiviral research 2019 01; 161 154-162	Majmudar, Hardik; Hao, Meng; Sankaranarayanan, Nehru Viji; Zanotti, Brian; Volin, Michael V; Desai, Umesh R; Tiwari, Vaibhav	2019			0.46
Polymeric fluorescent heparin as one-step FRET substrate of human heparanase.					
Carbohydrate polymers 2019 Feb 01; 205 385-391	Sistla, Jyothi C; Morla, Shravan; Alabbas, Al-Humaidi B; Kalathur, Ravi C; Sharon, Chetna; Patel, Bhaumik B; Desai, Umesh R	2019			1.11
Novel heparin mimetics reveal cooperativity between exosite 2 and sodium-binding site of thrombin.					
Thrombosis research 2018 May; 165 61-67	Abdel Aziz, May H; Desai, Umesh R	2018			0.33
A small group of sulfated benzofurans induces steady-state submaximal inhibition of thrombin.					
Bioorganic & medicinal chemistry letters 2018 Apr 01; 28 (6) 1101-1105	Afosah, Daniel K; Verespy 3rd, Stephen; Al-Horani, Rami A; Boothello, Rio S; Karuturi, Rajesh; Desai, Umesh R	2018			1.00
So you think computational approaches to understanding glycosaminoglycan-protein interactions are too dry and too rigid? Think again!					
Current opinion in structural biology 2018 Jun; 50 91-100	Sankaranarayanan, Nehru Viji; Nagarajan, Balaji; Desai, Umesh R	2018			3.80
A Synthetic, Small, Sulfated Agent Is a Promising Inhibitor of Chlamydia spp. Infection in vivo.					
Frontiers in microbiology 2018; 9 3269	Gallegos, Karen M; Taylor, Christopher R; Rabulinski, Daniel J; Del Toro, Rosalinda; Girgis, Danielle E; Jourha, Dapinder; Tiwari, Vaibhav; Desai, Umesh R; Ramsey, Kyle H View Less	2018			
Inhibition of Herpes Simplex Virus-1 Entry into Human Cells by Nonsaccharide Glycosaminoglycan Mimetics.					
ACS medicinal chemistry letters 2018 Aug 09; 9 (8) 797-802	Gangji, Rahaman Navaz; Sankaranarayanan, Nehru Viji; Elste, James; Al-Horani, Rami A; Afosah, Daniel K; Joshi, Rachel; Tiwari, Vaibhav; Desai, Umesh R	2018			1.17
Potent, Selective, Allosteric Inhibition of Human Plasmin by Sulfated Non-Saccharide Glycosaminoglycan Mimetics.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Journal of medicinal chemistry 2017 01 26; 60 (2) 641-657	Afosah, Daniel K; Al-Horani, Rami A; Sankaranarayanan, Nehru Viji; Desai, Umesh R	2017			1.41
Allosterism-based simultaneous, dual anticoagulant and antiplatelet action: allosteric inhibitor targeting the glycoprotein Ibα-binding and heparin-binding site of thrombin.					
Journal of thrombosis and haemostasis : JTH 2016 Apr; 14 (4) 828-38	Mehta, A Y; Mohammed, B M; Martin, E J; Brophy, D F; Gailani, D; Desai, U R	2016			1.27
Heparan sulfate hexasaccharide selectively inhibits cancer stem cells self-renewal by activating p38 MAP kinase.					
Oncotarget 2016 Dec 20; 7 (51) 84608-84622	Patel, Nirmita J; Sharon, Chetna; Baranwal, Somesh; Boothello, Rio S; Desai, Umesh R; Patel, Bhaumik B	2016			1.35
Allosteric Inhibition of Factor XIIIa. Non-Saccharide Glycosaminoglycan Mimetics, but Not Glycosaminoglycans, Exhibit Promising Inhibition Profile.					
PloS one 2016; 11 (7) e0160189	Al-Horani, Rami A; Karuturi, Rajesh; Lee, Michael; Afosah, Daniel K; Desai, Umesh R	2016			0.81
Broad Spectrum Anti-Influenza Agents by Inhibiting Self-Association of Matrix Protein 1.					
Scientific reports 2016 08 30; 6 32340	Mosier, Philip D; Chiang, Meng-Jung; Lin, Zhengshi; Gao, Yamei; Althufairi, Bashayer; Zhou, Qibing; View All	2016			0.42
Factor XIa inhibitors: A review of the patent literature.					
Expert opinion on therapeutic patents 2016; 26 (3) 323-45	Al-Horani, Rami A; Desai, Umesh R	2016			2.17
Allosteric Partial Inhibition of Monomeric Proteases. Sulfated Coumarins Induce Regulation, not just Inhibition, of Thrombin.					
Scientific reports 2016 Apr 07; 6 24043	Verespy 3rd, Stephen; Mehta, Akul Y; Afosah, Daniel; Al-Horani, Rami A; Desai, Umesh R	2016			1.31
Estimating glycosaminoglycan-protein interaction affinity: water dominates the specific antithrombin-heparin interaction.					
Glycobiology 2016 Oct; 26 (10) 1041-1047	Sarkar, Aurijit; Yu, Wenbo; Desai, Umesh R; MacKerell, Alexander D; Mosier, Philip D	2016			0.65
6-Hydroxyflavone and derivatives exhibit potent anti-inflammatory activity among mono-, di- and polyhydroxylated flavones in kidney mesangial cells.					
PloS one 2015; 10 (3) e0116409	Wang, Xing; Wang, Zhiwei; Sidhu, Preetpal Singh; Desai, Umesh R; Zhou, Qibing	2015			1.11
Investigation of the heparin-thrombin interaction by dynamic force spectroscopy.					
Biochimica et biophysica acta 2015 Jun; 1850 (6) 1099-106	Wang, Congzhou; Jin, Yingzi; Desai, Umesh R; Yadavalli, Vamsi K	2015			0.38
A Simple Method for Discovering Druggable, Specific Glycosaminoglycan-Protein Systems. Elucidation of Key Principles from Heparin/Heparan Sulfate-Binding Proteins.					
PloS one 2015; 10 (10) e0141127	Sarkar, Aurijit; Desai, Umesh R	2015			1.69
Glycosaminoglycan-protein interaction studies using fluorescence spectroscopy.					
Methods in molecular biology (Clifton, N.J.) 2015; 1229 335-53	Boothello, Rio S; Al-Horani, Rami A; Desai, Umesh R	2015			0.42
Studying glycosaminoglycan-protein interactions using capillary electrophoresis.					
Methods in molecular biology (Clifton, N.J.) 2015; 1229 355-75	Liang, Aiye; Desai, Umesh R	2015			0.07

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Designing "high-affinity, high-specificity" glycosaminoglycan sequences through computerized modeling.					
Methods in molecular biology (Clifton, N.J.) 2015; 1229 289-314	Sankaranarayanan, Nehru Viji; Sarkar, Aurijit; Desai, Umesh R; Mosier, Philip D	2015			iCite 0.62
Allosteric inhibition of factor XIa. Sulfated non-saccharide glycosaminoglycan mimetics as promising anticoagulants.					
Thrombosis research 2015 Aug; 136 (2) 379-87	Al-Horani, Rami A; Gailani, David; Desai, Umesh R	2015			iCite 1.55
An update on recent patents on thrombin inhibitors (2010 - 2013).					
Expert opinion on therapeutic patents 2014 Jan; 24 (1) 47-67	Mehta, Akul Y; Jin, Yingzi; Desai, Umesh R	2014			iCite 0.51
Sulfated caffeic acid dehydropolymer attenuates elastase and cigarette smoke extract-induced emphysema in rats: sustained activity and a need of pulmonary delivery.					
Lung 2014 Aug; 192 (4) 481-92	Saluja, Bhawana; Li, Hua; Desai, Umesh R; Voelkel, Norbert F; Sakagami, Masahiro	2014			iCite 0.34
Designing allosteric inhibitors of factor XIa. Lessons from the interactions of sulfated pentagalloylglucopyranosides.					
Journal of medicinal chemistry 2014 Jun 12; 57 (11) 4805-18	Al-Horani, Rami A; Desai, Umesh R	2014			iCite 1.71
Recent advances on plasmin inhibitors for the treatment of fibrinolysis-related disorders.					
Medicinal research reviews 2014 Nov; 34 (6) 1168-216	Al-Horani, Rami A; Desai, Umesh R	2014			iCite 2.09
Substantial non-electrostatic forces are needed to induce allosteric disruption of thrombin's active site through exosite 2.					
Biochemical and biophysical research communications 2014 Sep 26; 452 (3) 813-6	Mehta, Akul Y; Desai, Umesh R	2014			iCite 0.33
Discovery methodology for the development of direct factor VIIa inhibitors.					
Expert opinion on drug discovery 2014 Aug; 9 (8) 859-72	Henry, Brian L; Desai, Umesh R	2014			iCite 0.09
Synthetic, non-saccharide, glycosaminoglycan mimetics selectively target colon cancer stem cells.					
ACS chemical biology 2014 Aug 15; 9 (8) 1826-33	Patel, Nirmita J; Karuturi, Rajesh; Al-Horani, Rami A; Baranwal, Somesh; Patel, Jagrut; Desai, Umesh R; Patel, Bhaumik B	2014			iCite 1.14
A simple, general approach of allosteric coagulation enzyme inhibition through monosulfated hydrophobic scaffolds.					
Bioorganic & medicinal chemistry letters 2014 Dec 15; 24 (24) 5716-5720	Sidhu, Preetpal Singh; Zhou, Qibing; Desai, Umesh R	2014			iCite 0.14
Targeting the GPIbα binding site of thrombin to simultaneously induce dual anticoagulant and antiplatelet effects.					
Journal of medicinal chemistry 2014 Apr 10; 57 (7) 3030-9	Mehta, Akul Y; Thakkar, Jay N; Mohammed, Bassem M; Martin, Erika J; Brophy, Donald F; Kishimoto, Takao; Desai, Umesh R	2014			iCite 0.76
Allosteric inhibition of human factor XIa: discovery of monosulfated benzofurans as a class of promising inhibitors.					
Journal of medicinal chemistry 2014 Apr 24; 57 (8) 3559-69	Argade, Malaika D; Mehta, Akul Y; Sarkar, Aurijit; Desai, Umesh R	2014			iCite 0.86
Sulfated low molecular weight lignins, allosteric inhibitors of coagulation proteinases via the heparin binding site, significantly alter the active site of thrombin and factor xa compared to heparin.					
Thrombosis research 2014 Nov; 134 (5) 1123-9	Henry, Brian L; Desai, Umesh R	2014			iCite 0.58
On scaffold hopping: challenges in the discovery of sulfated small molecules as mimetics of glycosaminoglycans.					
Bioorganic & medicinal chemistry letters 2013 Jan 01; 23 (1) 355-9	Sidhu, Preetpal S; Mosier, Philip D; Zhou, Qibing; Desai, Umesh R	2013			iCite 0.36
Sulfated pentagalloylglucoside is a potent, allosteric, and selective inhibitor of factor XIa.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Journal of medicinal chemistry 2013 Feb 14; 56 (3) 867-78	Al-Horani, Rami A; Ponnusamy, Pooja; Mehta, Akul Y; Gailani, David; Desai, Umesh R	2013			2.60
Novel low molecular weight lignins as potential anti-emphysema agents: In vitro triple inhibitory activity against elastase, oxidation and inflammation.					
Pulmonary pharmacology & therapeutics 2013 Apr; 26 (2) 296-304	Saluja, Bhawana; Thakkar, Jay N; Li, Hua; Desai, Umesh R; Sakagami, Masahiro	2013			0.87
Designing allosteric regulators of thrombin. Exosite 2 features multiple subsites that can be targeted by sulfated small molecules for inducing inhibition.					
Journal of medicinal chemistry 2013 Jun 27; 56 (12) 5059-70	Sidhu, Preetpal Singh; Abdel Aziz, May H; Sarkar, Aurijit; Mehta, Akul Y; Zhou, Qibing; Desai, Umesh R	2013			1.77
Discovery of allosteric modulators of factor XIa by targeting hydrophobic domains adjacent to its heparin-binding site.					
Journal of medicinal chemistry 2013 Mar 28; 56 (6) 2415-28	Karuturi, Rajesh; Al-Horani, Rami A; Mehta, Shrenik C; Gailani, David; Desai, Umesh R	2013			1.36
The promise of sulfated synthetic small molecules as modulators of glycosaminoglycan function.					
Future medicinal chemistry 2013 Aug; 5 (12) 1363-6	Desai, Umesh R	2013			1.01
Dynamic affinity chromatography in the separation of sulfated lignins binding to thrombin.					
Journal of chromatography. B, Analytical technologies in the biomedical and life sciences 2012 Nov 01; 908 45-51	Liang, Aiye; Thakkar, Jay N; Hindle, Michael; Desai, Umesh R	2012			
On the specificity of heparin/heparan sulfate binding to proteins. Anion-binding sites on antithrombin and thrombin are fundamentally different.					
PloS one 2012; 7 (11) e48632	Mosier, Philip D; Krishnasamy, Chandravel; Kellogg, Glen E; Desai, Umesh R	2012			1.53
Potent direct inhibitors of factor Xa based on the tetrahydroisoquinoline scaffold.					
European journal of medicinal chemistry 2012 Aug; 54 771-83	Al-Horani, Rami A; Mehta, Akul Y; Desai, Umesh R	2012			0.42
Sulfated, low molecular weight lignins inhibit a select group of heparin-binding serine proteases.					
Biochemical and biophysical research communications 2012 Jan 6; 417 (1) 382-6	Henry, Brian L; Thakkar, Jay N; Liang, Aiye; Desai, Umesh R	2012			0.70
Discovery of novel sulfonated small molecules that inhibit vascular tube formation.					
Bioorganic & medicinal chemistry letters 2012 Jul 1; 22 (13) 4467-70	Raman, Karthik; Karuturi, Rajesh; Swarup, Vimal P; Desai, Umesh R; Kuberan, Balagurunathan	2012			0.59
Electronically Rich N-Substituted Tetrahydroisoquinoline 3-Carboxylic Acid Esters: Concise Synthesis and Conformational Studies.					
Tetrahedron 2012 Feb 25; 68 (8) 2027-2040	Al-Horani, Rami A; Desai, Umesh R	2012			0.30
Designing allosteric regulators of thrombin. Monosulfated benzofuran dimers selectively interact with Arg173 of exosite 2 to induce inhibition.					
Journal of medicinal chemistry 2012 Aug 9; 55 (15) 6888-97	Abdel Aziz, May H; Sidhu, Preetpal Singh; Liang, Aiye; Kim, Ji Yeong; Mosier, Philip D; Zhou, Qibing; Farrell, David H; Desai, Umesh R	2012			1.26
Designing nonsaccharide, allosteric activators of antithrombin for accelerated inhibition of factor Xa.					
Journal of medicinal chemistry 2011 Sep 8; 54 (17) 6125-38	Al-Horani, Rami A; Liang, Aiye; Desai, Umesh R	2011			1.08
Linear polyalkylamines as fingerprinting agents in capillary electrophoresis of low-molecular-weight heparins and glycosaminoglycans.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Electrophoresis 2011 Nov; 32 (21) 3070-7	King, J Timothy; Desai, Umesh R	2011			0.13
Rational design of potent, small, synthetic allosteric inhibitors of thrombin.					
Journal of medicinal chemistry 2011 Aug 11; 54 (15) 5522-31	Sidhu, Preetpal Singh; Liang, Aiye; Mehta, Akul Y; Abdel Aziz, May H; Zhou, Qibing; Desai, Umesh R	2011			1.54
Serpin-glycosaminoglycan interactions.					
Methods in enzymology 2011; 501 105-37	Rein, Chantelle M; Desai, Umesh R; Church, Frank C	2011			0.94
Interaction of thrombin with sucrose octasulfate.					
Biochemistry 2011 Aug 16; 50 (32) 6973-82	Desai, Bijoy J; Boothello, Rio S; Mehta, Akul Y; Scarsdale, J Neel; Wright, H Tonie; Desai, Umesh R	2011			0.62
Identification of the site of binding of sulfated, low molecular weight lignins on thrombin.					
Biochemical and biophysical research communications 2011 Sep 23; 413 (2) 348-52	Abdel Aziz, May H; Mosier, Philip D; Desai, Umesh R	2011			0.60
Study of physico-chemical properties of novel highly sulfated, aromatic, mimetics of heparin and heparan sulfate.					
Journal of pharmaceutical sciences 2010 Mar; 99 (3) 1207-16	Liang, Aiye; Thakkar, Jay N; Desai, Umesh R	2010			0.65
Sulfated, low-molecular-weight lignins are potent inhibitorsof plasmin, in addition to thrombin and factor Xa: Novel opportunity for controlling complex pathologies.					
Thrombosis and haemostasis 2010 Mar; 103 (3) 507-15	Henry, Brian L; Abdel Aziz, May; Zhou, Qibing; Desai, Umesh R	2010			0.83
Chemical Sulfation of Small Molecules - Advances and Challenges.					
Tetrahedron 2010 Apr 17; 66 (16) 2907-2918	Al-Horani, Rami A; Desai, Umesh R	2010			2.66
Nonsulfated, cinnamic acid-based lignins are potent antagonists of HSV-1 entry into cells.					
Biomacromolecules 2010 May 10; 11 (5) 1412-6	Thakkar, Jay N; Tiwari, Vaibhav; Desai, Umesh R	2010			0.68
Interaction of antithrombin with sulfated, low molecular weight lignins: opportunities for potent, selective modulation of antithrombin function.					
The Journal of biological chemistry 2009 Jul 31; 284 (31) 20897-908	Henry, Brian L; Connell, Justin; Liang, Aiye; Krishnasamy, Chandravel; Desai, Umesh R	2009			0.95
First steps in the direction of synthetic, allosteric, direct inhibitors of thrombin and factor Xa.					
Bioorganic & medicinal chemistry letters 2009 Aug 1; 19 (15) 4126-9	Verghese, Jenson; Liang, Aiye; Sidhu, Preet Pal Singh; Hindle, Michael; Zhou, Qibing; Desai, Umesh R	2009			0.99

Patents

No Patents information available for 5R01HL090586-08

Outcomes

The Project Outcomes shown here are displayed verbatim as submitted by the Principal Investigator (PI) for this award. Any opinions, findings, and conclusions or recommendations expressed are those of the PI and do not necessarily reflect the views of the National Institutes of Health. NIH has not endorsed the content below.

1. For the first time, we developed the proof-of-principle that allosteric anticoagulant and antiplatelet agents can be designed by

exploiting the interaction of coagulation factors with glycosaminoglycans. We designed several small molecule agents as well polymeric agents. The coagulation factors can be thrombin, factor XIa, factor XIIIa, and plasmin. These small molecule agents for the first generation agents, which will form the foundation for the design of advanced agents that display better safety and efficacy profiles. 2. We realized this promising concept in the form of SPGG and SCI; two small molecule, allosteric inhibitors that reduce clotting, without any major bleeding complications in mice. The bleeding phenomenon besets current anticoagulants in use today. 3 We also showed that other proteins can be similarly targeted including human neutrophil elastase, which plays a role in inflammatory processes. 4. For the first time, we showed that regulation, not just inhibition, of thrombin can be induced by small molecules. 5. For the first time, we showed that rigorous use of computational techniques can lead to discovery of unique and rare GAG sequences that can selectively target GAG-binding protein(s), which may lead to synthetic agents as chemical biology or drug discovery tools. 6. This work led to the synthesis of a large library of sulfated small molecules. This large library was used to screen against various proteins and cells involved in diseases other than thrombosis. This work was accomplished with the aid of collaborators and yielded a rich trove of knowledge. Our work established that some of the members of the library inhibited cancer stem cells, human neutrophil elastase, glycoprotein D, and matrix metalloproteinase-8, which play roles in cancer, cystic fibrosis and viral infection. 7. This work also involved performing extensive studies on how computational approaches could be developed to understand the biosynthesis of highly sulfated biomolecules.This greatly enhanced the discovery of antagonists of coagulation and fibrinolytic agents. 8. We educated scientists on experimental methods and protocols for studying the interaction of glycosaminoglycans with proteins using fluorescence spectroscopy, capillary electrophoresis, and molecular modeling. We also educated experts on the technique of computational modeling and its use in the discovery of agents that mimic the interaction of glycosaminoglycans with proteins.

Clinical Studies

No Clinical Studies information available for 5R01HL090586-08

News and More

Related News Releases

No news release information available for 5R01HL090586-08

History

Total project funding amount for 9 projects is $3,939,023*

Export

** Only NIH, CDC and FDA funding data*

Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Cos IC
Allosteric Inhibitors of Coagulation Proteases							
5R01HL090586-08		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2018	NHLBI	NHLBI	$512,380
Allosteric Inhibitors of Coagulation Proteases							
3R01HL090586-08S1		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2018	NHLBI	NIA	$394,375
Allosteric Inhibitors of Coagulation Proteases							
5R01HL090586-07		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2017	NHLBI	NHLBI	$510,826
Allosteric Inhibitors of Coagulation Proteases							
5R01HL090586-06		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2016	NHLBI	NHLBI	$509,289
Allosteric Inhibitors of Coagulation Proteases							
2R01HL090586-05		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2015	NHLBI	NHLBI	$520,890

Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
Dual Direct Inhibitors of Thrombin and Factor Xa							
5R01HL090586-04		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2012	NHLBI	NHLBI	$370,013
Dual Direct Inhibitors of Thrombin and Factor Xa							
5R01HL090586-03		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2011	NHLBI	NHLBI	$373,750
Dual Direct Inhibitors of Thrombin and Factor Xa							
5R01HL090586-02		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2010	NHLBI	NHLBI	$373,750
Dual Direct Inhibitors of Thrombin and Factor Xa							
1R01HL090586-01A2		DESAI, UMESH RAMANLAL	VIRGINIA COMMONWEALTH UNIVERSITY	2009	NHLBI	NHLBI	$373,750

Similar Projects

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Tot Cost b
Prevention of postpartum hemorrhage: identifying pregnant women at risk and determining the safe and effective use of tranexamic acid using state-of-the-art pharmacokinetic/pharmacodynamics modeling								
297	5K23HL141640-04		AHMADZIA, HOMA KHORRAMI	GEORGE WASHINGTON UNIVERSITY	2022	NHLBI	NHLBI	$155,174
Pediatric ECMO Anticoagulation Collaborative (PEACE) Consensus								
266	1R13HD104432-01		ALEXANDER, PETA-MAREE ANN MUSZYNSKI, JENNIFER	BOSTON CHILDREN'S HOSPITAL	2021	NICHD	NICHD	$74,880
Platelet alphaIIbbeta3 activation and its therapeutic targeting								
277	5R01HL141366-04		ARNAOUT, M. AMIN	MASSACHUSETTS GENERAL HOSPITAL	2022	NHLBI	NHLBI	$533,504
The interaction of basal risk, pharmacologic ovulation induction, pregnancy and delivery on hemostatic balance								
442	5R33HL141787-05		BERNSTEIN, IRA MARK ORFEO, THOMAS	UNIVERSITY OF VERMONT & ST AGRIC COLLEGE	2022	NHLBI	NHLBI	$730,424
Investigating structure activity relationships in autoprocessing by HIV-1 protease								
265	5R21AI157871-02		BOLON, DANIEL N	UNIV OF MASSACHUSETTS MED SCH WORCESTER	2022	NIAID	NIAID	$209,375
Impact of Clot Histological and Physical Properties on Revascularization Strategies in Acute Ischaemic Stroke								
237	5R01NS105853-05		BRINJIKJI, WALEED	MAYO CLINIC ROCHESTER	2022	NINDS	NINDS	$347,813
Colon cancer prevention with non-systemic PDE5 inhibitors								
297	1R41CA272072-01		BROWNING, DARREN D.	TARGUT BIOTECHNOLOGIES INC.	2022	NCI	NCI	$191,738
Oxadiazole Inhibitors of Non-Stop Ribosome Rescue to treat MDR Neisseria gonorrhoeae								
276	5R01AI132276-05		BUTLER, MICHELLE M.	MICROBIOTIX, INC	2021	NIAID	NIAID	$1,052,607
Clot-Targeted Antithrombotics for Venous Thromboprophylaxis								
867	5R01HL147855-04		CHAIKOF, ELLIOT	BETH ISRAEL DEACONESS MEDICAL CENTER	2022	NHLBI	NHLBI	$465,803

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
Small Molecule Protease Inhibitors against MERS-CoV								
249	5R01AI130092-05		CHANG, KYEONG-OK GROUTAS, WILLIAM CONSTANTINE PERLMAN, STANLEY	KANSAS STATE UNIVERSITY	2022	NIAID	NIAID	$728,584
A general, virus-free platform to rapidly map SARS-CoV-2 drug resistance								
243	5R21AI162662-02		CHOUDHARY, AMIT LIAU, BRIAN	BROAD INSTITUTE, INC.	2022	NIAID	NIAID	$195,000
Microvascular Thrombosis in systemic inflammation								
224	5R01GM112806-08		CRUZ, MIGUEL ANGEL NGUYEN, TRUNG CHANH VIJAYAN, K. VINOD	BAYLOR COLLEGE OF MEDICINE	2022	NIGMS	NIGMS	$375,236
Regulation of Normal and Asthmatic Lung Function by G-Protein-Coupled Receptors								
287	1ZIAAI000939-19		DRUEY, KIRK M	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$411,004
Use and outcomes of anticoagulants for the treatment and prevention of thrombosis among hospitalized patients								
548	5K24HL141354-05		FANG, MARGARET C.	UNIVERSITY OF CALIFORNIA, SAN FRANCISCO	2022	NHLBI	NHLBI	$121,834
Biophysics of Macromolecular Complexes								
349	1ZIADK033007-18		FELSENFELD, GARY	NATIONAL INSTITUTE OF DIABETES AND DIGESTIVE AND KIDNEY DISEASES	2022	NIDDK	NIDDK	$396,360
Biomarkers and mechanisms in cancer associated thrombosis								
292	5U01HL143365-05		FLAUMENHAFT, ROBERT C CHAIKOF, ELLIOT ZWICKER, JEFFREY	BETH ISRAEL DEACONESS MEDICAL CENTER	2022	NHLBI	NHLBI	$829,929
Targeting the Plasminogen Activation System to Limit Pancreatic Cancer Progression and Associated Thrombosis								
286	5U01HL143403-05		FLICK, MATTHEW J FISHEL, MELISSA L HAN, BUMSOO WOLBERG, ALISA S.	UNIV OF NORTH CAROLINA CHAPEL HILL	2022	NHLBI	NHLBI	$834,158
Thrombin-dependent mechanisms of pancreatic ductal adenocarcinoma disease								
404	5R01CA211098-06		FLICK, MATTHEW J	UNIV OF NORTH CAROLINA CHAPEL HILL	2022	NCI	NCI	$403,147
Safety and Effectiveness of Left Atrial Appendage Closure in Atrial Fibrillation (SAFELY-AF)								
226	5R01HL142765-04		FREEMAN, JAMES	YALE UNIVERSITY	2022	NHLBI	NHLBI	$701,023
A novel clot containing system for improved neurothrombectomy outcomes								
228	1R41NS129430-01		GAMBA, JORGE	XTRACT MEDICAL, INC.	2022	NINDS	NINDS	$485,902
Developing non-anticoagulant recombinant heparin for treating the COPD Pathogenic Triad								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
696	1R43HL162135-01		GLASS, CHARLES ALEXANDER THACKER, BRYAN EDWARD	TEGA THERAPEUTICS, INC.	2022	NHLBI	NHLBI	$321,866
KidsDOTT-CCC								
401	5U01HL130048-05		GOLDENBERG, NEIL A	JOHNS HOPKINS UNIVERSITY	2021	NHLBI	NHLBI	$405,447
Role of acid sphingomyelinase in the modulation of coagulation after traumatic brain injury								
270	5R01GM124156-05		GOODMAN, MICHAEL	UNIVERSITY OF CINCINNATI	2022	NIGMS	NIGMS	$304,950
The Roles of Complement Activation vs. Shiga Toxin Binding to Endothelium in eHUS.								
266	5R01DK117317-04		GRABOWSKI, ERIC FRANKLIN	MASSACHUSETTS GENERAL HOSPITAL	2022	NIDDK	NIDDK	$408,176
Multiscale Modeling of Clotting Risk in Atrial Fibrillation								
358	5U01HL143336-05		GRIFFITH, BOYCE EUGENE	UNIV OF NORTH CAROLINA CHAPEL HILL	2022	NHLBI	NHLBI	$545,784
Impact of steady state Cobicistat and Darunavir/Cobicistat And on the Pharmacokinetics And Pharmacodynamics of Oral Anticoagulants (Rivaroxaban, Apixaban) In Healthy Volunteers (CLOTRX)								
572	1ZIACL006017-04		HADIGAN, COLLEEN	CLINICAL CENTER	2022	CLC		
Cerebrovascular involvement in Alzheimer's Disease: PAR4 Antagonism								
251	1RF1AG068623-01A1		HAMM, HEIDI E MEILER, JENS	VANDERBILT UNIVERSITY	2021	NIA	NIA	$2,185,927
Clinical Importance of Drug-Drug Interactions								
359	5R01AG025152-15		HENNESSY, SEAN LI, LANG	UNIVERSITY OF PENNSYLVANIA	2021	NIA	NIA	$628,764
Multiplex Small Molecule Discovery to Identify Broad-Acting Viral Protease Inhibitors								
240	1U19AI171401-01	8299	HO, DAVID D	HACKENSACK UNIVERSITY MEDICAL CENTER	2022	NIAID	NIAID	*$3,545,618*
Biomarkers for 12-lipoxygenase inhibition as a therapeutic intervention for heparin-induced thrombocytopenia and thrombosis (HIT/T)								
552	5R21TR003185-02		HOLINSTAT, MICHAEL ALLAN	UNIVERSITY OF MICHIGAN AT ANN ARBOR	2022	NCATS	NCATS	$233,600
Accelerated Low Dose Thrombolytic Catheter Directed Sonothrombolysis								
494	5R01HL141752-04		HOSSACK, JOHN A	UNIVERSITY OF VIRGINIA	2021	NHLBI	NHLBI	$517,715
Evolving New Glycosaminoglycan Mimetics								
623	5R01GM127920-04		HSIEH-WILSON, LINDA C	CALIFORNIA INSTITUTE OF TECHNOLOGY	2021	NIGMS	NIGMS	$384,183
Novel α2-Antiplasmin Inactivation for Lysis of Intravascular Thrombi (NAIL-IT) Trial								
290	5R44HL158376-02		JEONG, SUN YONG REED, GUY L TAPSON, VICTOR F.	TRANSLATIONAL SCIENCES, INC.	2022	NHLBI	NHLBI	$1,408,690
Forward viewing catheter-delivered microbubble enhanced sonothrombolysis (FV-CAMUS)								
352	5R01HL141967-04		JIANG, XIAONING	NORTH CAROLINA STATE UNIVERSITY RALEIGH	2021	NHLBI	NHLBI	$522,382
Small molecule mimetics of Humanin that normalize neuronal p-Akt as novel therapeutics for AD								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
263	1RF1AG068116-01A1		JOHN, VARGHESE	UNIVERSITY OF CALIFORNIA LOS ANGELES	2021	NIA	NIA	$1,817,118
Unraveling Cytotoxic and Thrombotic Signals in COVID-19								
228	1ZIAHL006263-02		KANTHI, YOGEN	NATIONAL HEART, LUNG, AND BLOOD INSTITUTE	2022	NHLBI	NHLBI	$963,672
Human Translational Studies of Vascular Thrombosis and Inflammation								
281	1ZIAHL006262-02		KANTHI, YOGEN	NATIONAL HEART, LUNG, AND BLOOD INSTITUTE	2022	NHLBI	NHLBI	$963,672
Inhibition of CaVα-β interaction with orally available small organic molecules for chronic pain								
231	1R41CA257845-01A1		KHANNA, RAJESH	REGULONIX, LLC	2021	NCI	NCI	$400,000
First-in-class ETS inhibitor TK216: Translational Biology and Oral Dosage Form Development								
231	5R44CA261557-02		KRISHNAN, RAJESH	ONCTERNAL THERAPEUTICS, INC.	2022	NCI	NCI	$1,023,231
Hemostasis and Thrombosis: Chemistry, Biology and Physiology								
585	5P01HL139420-05		KRISHNASWAMY, SRIRAM	CHILDREN'S HOSP OF PHILADELPHIA	2022	NHLBI	NHLBI	$1,637,113
Protease Stable N-Terminally Modified Therapeutic Peptides								
293	1R41DK131842-01A1		KUMAR, KRISHNA	VELUM, INC.	2022	NIDDK	NIDDK	$367,389
Vaccines and Therapeutics for Anthrax								
227	1ZIAAI000929-20		LEPPLA, STEPHEN	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$770,304
Optimization of AAV vector to deliver FVa for hemophilia with inhibitors								
344	5R01HL144661-04		LI, CHENGWEN NICHOLS, TIMOTHY CHARLES	UNIV OF NORTH CAROLINA CHAPEL HILL	2022	NHLBI	NHLBI	$625,580
High throughput screening of orthosteric inhibitors of flavivirus protease								
296	5R01AI131669-05		LI, HONGMIN	UNIVERSITY OF ARIZONA	2022	NIAID	NIAID	$550,021
Structural and Functional Basis of the Vitamin K Cycle								
235	5R01HL121718-09		LI, WEIKAI	WASHINGTON UNIVERSITY	2022	NHLBI	NHLBI	$393,750
Inflammasome Activation Triggers Systemic Coagulation in Sepsis								
328	7R01HL142640-04		LI, ZHENYU WEI, YINAN	TEXAS A&M UNIVERSITY HEALTH SCIENCE CTR	2022	NHLBI	NHLBI	$429,469
Development of selective and potent protease inhibitors for corona and other pandemic viruses								
337	1U19AI171421-01	8526	LIN, MICHAEL Z.	STANFORD UNIVERSITY	2022	NIAID	NIAID	*$2,897,311*
Novel, Safe, Efficacious Heparin Reversal								
330	1R43HL165961-01		LIPKA, ELKE	TSRL, INC.	2022	NHLBI	NHLBI	$394,392
Tumor-promoting functions of TMPRSS13 in breast cancer progression								
280	5R01CA222359-05		LIST, KARIN	WAYNE STATE UNIVERSITY	2022	NCI	NCI	$309,573
Biochemical and Structural Studies of Viral Proteases								
480	1ZIADK075166-01		LOUIS, JOHN	NATIONAL INSTITUTE OF DIABETES AND DIGESTIVE AND KIDNEY DISEASES	2022	NIDDK	NIDDK	$695,808
Small Molecule E6 Inhibitors to Treat Oropharyngeal Cancers Caused by HPV Infections								

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267	1R43DE031495-01A1		LU, ZHIJIAN ANDROPHY, ELLIOT J.	KOVINA THERAPEUTICS INC.	2022	NIDCR	NIDCR	$275,766
DNA Replication and Repair								
282	1ZIABC012086-01		MACHIDA, YUICHI	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$1,181,073
Prevention of HIV-1 transmission by small-molecule CD4-mimetic entry inhibitors								
301	5R01AI134494-05		MADANI, NAVID	DANA-FARBER CANCER INST	2021	NIAID	NIAID	$651,658
AtRial Cardiopathy and Antithrombotic Drugs In prevention After cryptogenicstroke (ARCADIA)								
248	5U01NS095869-05		MARSHALL, RANDOLPH S BRODERICK, JOSEPH PAUL KAMEL, HOOMAN LONGSTRETH, WT MEINZER, CAITLYN N TIRSCHWELL, DAVID L	COLUMBIA UNIVERSITY HEALTH SCIENCES	2022	NINDS	NINDS	$2,310,548
Characterization of an isoform specific anticoagulant function of TFPI-alpha								
452	5R01HL068835-16		MAST, ALAN E	VERSITI WISCONSIN, INC.	2022	NHLBI	NHLBI	$745,219
Characterization of Coagulation Factor-platelet Interactions: Role of FXI								
399	5R01HL101972-13		MCCARTY, OWEN J	OREGON HEALTH & SCIENCE UNIVERSITY	2022	NHLBI	NHLBI	$701,086
Novel approaches to improve prediction of cancer-associated thrombosis								
283	5U01HL143402-05		MCCRAE, KEITH R. KHORANA, ALOK A	CLEVELAND CLINIC LERNER COM-CWRU	2022	NHLBI	NHLBI	$957,703
Nanoparticle- Infused Radiopaque Absorbable Medical Device								
315	5R01HL141831-05		MELANCON, MARITES PASUELO	UNIVERSITY OF TX MD ANDERSON CAN CTR	2022	NHLBI	NHLBI	$386,308
UTMB-Novartis Alliance for Pandemic Preparedness								
224	1U19AI171413-01		MENACHERY, VINEET D DIAGANA, THIERRY	UNIVERSITY OF TEXAS MED BR GALVESTON	2022	NIAID	NIAID	$56,163,625
Integrated Coagulation Sensing to Predict Hemorrhage and Guide Transfusions								
411	5R01HL142272-04		NADKARNI, SEEMANTINI K	MASSACHUSETTS GENERAL HOSPITAL	2021	NHLBI	NHLBI	$790,843
ASK1 a novel regulator of platelet function								
243	5R01HL113188-08		NAIK, ULHAS P	THOMAS JEFFERSON UNIVERSITY	2022	NHLBI	NHLBI	$477,012
Endogenous suppression of integrin signaling								
222	5R01HL119374-09		NAIK, ULHAS P	THOMAS JEFFERSON UNIVERSITY	2022	NHLBI	NHLBI	$580,289
Increasing efficiency of sdFv production in a tobacco-based system for synthetic platelet design								
232	1R41HL162242-01		NANDI, SEEMA	SELSYM BIOTECH, INC.	2022	NHLBI	NHLBI	$258,501

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
Tailoring Structures of Sulfated Oligosaccharides for Modulating Heparanase Activity								
421	5R01GM098285-09		NGUYEN, HIEN M	WAYNE STATE UNIVERSITY	2022	NIGMS	NIGMS	$346,594
HIV Integrase Modeling and Computer-Aided Inhibitor and Microbicide Development								
239	1ZIABC010524-20		NICKLAUS, MARC	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$36,118
Digital Microfluidics - Minimizing Blood Volume for Pediatric Coagulation Screening								
354	5R44HL125484-06		PAMULA, VAMSEE K.	BAEBIES, INC.	2022	NHLBI	NHLBI	$995,038
Activity-based protein profiling of the human rhomboid proteases for inhibitor discovery and enzyme characterization								
226	1R15GM146210-01		PARSONS, WILLIAM HAZEN	OBERLIN COLLEGE	2022	NIGMS	NIGMS	$372,264
Crystallographic studies of macromolecular structures								
317	1ZICES102645-14		PEDERSEN, LARS	NATIONAL INSTITUTE OF ENVIRONMENTAL HEALTH SCIENCES	2022	NIEHS	NIEHS	$1,612,325
Accelerated development of advanced leads against SARS-CoV-2 and other pandemic viruses								
318	1U19AI171401-01	8296	PERLIN, DAVID S	HACKENSACK UNIVERSITY MEDICAL CENTER	2022	NIAID	NIAID	*$3,885,770*
Platelet-hyaluronan interactions as regulators of inflammation and thrombosis								
232	5R00HL135265-05		PETREY, AARON CHRISTOPHER	UNIVERSITY OF UTAH	2022	NHLBI	NHLBI	$249,000
Role of Secretoglobin 1A1 Protein in COVID-19-induced Thrombosis								
260	1R43HL158368-01		PILON, APRILE L	APCBIO INNOVATIONS, INC.	2021	NHLBI	NHLBI	$256,467
Vector-Borne Diseases: Molecular Mechanisms in Vector-Host Interactions								
403	1ZIAAI001012-16		RIBEIRO, JOSE	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$517,844
A Novel Glycosaminoglycan-Based Therapeutic for Chronic Rhinosinusitis								
341	5R44AI126987-06		SAVAGE, JUSTIN RODNEY	GLYCOMIRA, LLC	2022	NIAID	NIAID	$976,018
Protease/PAR2/TRPV4 Axis and Oral Cancer Pain								
221	5R01DE026806-05		SCHMIDT, BRIAN L BUNNETT, NIGEL W	NEW YORK UNIVERSITY	2022	NIDCR	NIDCR	$707,316
A Study to Evaluate the Effects of fixed dose Flavonoid Isoquercetin on thrombo-inflammatory biomarkers in subjects with stable Sickle Cell Disease								
673	1ZIAHL006256-02		SHET, ARUN	NATIONAL HEART, LUNG, AND BLOOD INSTITUTE	2022	NHLBI	NHLBI	$556,140
Venous Thrombosis Biomarkers in Sickle Cell Disease and Sickle Cell Trait								
352	1ZIAHL006255-03		SHET, ARUN	NATIONAL HEART, LUNG, AND BLOOD INSTITUTE	2022	NHLBI	NHLBI	$202,909
Platelet mass microarchitecture as a regulator of thrombin production								
621	5R21HL153946-02		SINNO, TALID TOMAIUOLO, MAURIZIO	UNIVERSITY OF PENNSYLVANIA	2022	NHLBI	NHLBI	$203,125
Functional Characterization of the Schistosome Tegument								
269	2R56AI056273-16A1		SKELLY, PATRICK J	TUFTS UNIVERSITY BOSTON	2021	NIAID	NIAID	$412,500

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
Developing a new drug for treating myocardial ischemia/reperfusion injury								
309	5R44HL156560-02		SKIDGEL, RANDAL A	DUPAGE MEDICAL TECHNOLOGY, INC.	2022	NHLBI	NHLBI	$1,141,737
Novel small-molecule inhibitors of SARS-CoV-2 protease								
279	1R21AI159323-01		SONG, YONGCHENG	BAYLOR COLLEGE OF MEDICINE	2021	NIAID	NIAID	$440,000
Inhibition of coronavirus release and mitigate COVID-19 pathogenesis								
430	1ZIAAI001263-03		SUN, PETER D.	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$171,318
Development of ALG-097431 as a broad-spectrum 3CL pro inhibitor								
242	1U19AI171401-01	8295	SYMONS, JULIAN A	HACKENSACK UNIVERSITY MEDICAL CENTER	2022	NIAID	NIAID	*$4,228,031*
Direct Detection and Characterization of Thrombosis In Vivo								
498	5R01HL147147-04		TARANTAL, ALICE F SULLENGER, BRUCE ALAN	UNIVERSITY OF CALIFORNIA AT DAVIS	2022	NHLBI	NHLBI	$734,488
Tick saliva and its importance for tick feeding and pathogen transmission								
587	1ZIAAI001337-02		TIRLONI, LUCAS	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$1,423,607
Factor XII Inhibitor for Surface Initiated Thrombosis								
980	5R44HL126235-06		TUCKER, ERIK IAN LORENTZ, CHRISTINA U	ARONORA, INC.	2022	NHLBI	NHLBI	$953,492
Discovery of new antiviral methylase, protease, and helicase inhibitors of corona-, flavi-, and alphaviruses								
365	1U19AI171401-01	8297	TUSCHL, THOMAS	HACKENSACK UNIVERSITY MEDICAL CENTER	2022	NIAID	NIAID	*$5,701,101*
Systematic multidisciplinary approach to study traumatic bleeding as a complex structural and biomechanical problem								
471	5R00HL148646-04		TUTWILER, VALERIE	RUTGERS, THE STATE UNIV OF N.J.	2022	NHLBI	NHLBI	$249,000
Structural Investigation of human factor IX zymogen activation by TF.VIIa and fXIa:fXIa dimer complexes								
302	1R15HL150722-01		VENKATESWARLU, DIVI	NORTH CAROLINA AGRI & TECH ST UNIV	2020	NHLBI	NHLBI	$323,262
Heparan Sulfate 3-O-Sulfation in Transcellular Propagation of Tauopathy in Alzheimer's Disease								
311	1RF1AG069039-01		WANG, CHUNYU LINHARDT, ROBERT J LIU, JIAN WANG, LIANCHUN	RENSSELAER POLYTECHNIC INSTITUTE	2020	NIA	NIA	$2,595,120
High Throughput Screening of Inhibitors Targeting the Enterovirus A71 and D68 2A proteases								
300	5R01AI147325-05		WANG, JUN	RUTGERS, THE STATE UNIV OF N.J.	2022	NIAID	NIAID	$409,619
Translation Studie of the Planet Specific Receptor Trem Like Transcript (TLT)								
306	5R01HL090933-09		WASHINGTON, A. VALANCE	OAKLAND UNIVERSITY	2022	NHLBI	NHLBI	$366,835
PET fibrin imaging of DVT and PE								

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277	5R01HL141900-04		WINKLER, TILO	MASSACHUSETTS GENERAL HOSPITAL	2021	NHLBI	NHLBI	$379,125
Protein Structure								
236	1ZIABC010348-23		WLODAWER, ALEXANDER	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$1,923,299
Cognitive Dysfunction in Atrial Fibrillation								
237	5R03AG067207-02		WOOD, KATHRYN ALICE	EMORY UNIVERSITY	2022	NIA	NIA	$78,250
Mechanism of Inhibition of Entry Inhibitors against SARS-CoVs								
248	1ZIABC011972-03		XIA, DI S	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$111,994
Balancing Risks and Benefits of Direct Oral Anticoagulants in Older Ischemic Stroke Patients with Atrial Fibrillation								
536	7R01AG066672-03		XIAN, YING	UT SOUTHWESTERN MEDICAL CENTER	2022	NIA	NIA	$493,248
Small molecule inhibitors and degraders of picornavirus 2A proteases as direct-acting antivirals								
250	1U19AI171421-01	8525	YANG, PRISCILLA LI-NING	STANFORD UNIVERSITY	2022	NIAID	NIAID	*$3,728,844*
Study of and development of antiviral drugs against SARS-CoV2 protease								
361	1ZIABC011945-03		YARCHOAN, ROBERT	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$167,849
Discovery of novel flavivirus inhibitors								
283	1U19AI171413-01	8446	YOKOKAWA, FUMIAKI	UNIVERSITY OF TEXAS MED BR GALVESTON	2022	NIAID	NIAID	*$2,677,020*
Thrombosis and Bleeding Risk Assessment in Medical Inpatients								
301	5R01HL141290-04		ZAKAI, NEIL ADRIAN	UNIVERSITY OF VERMONT & ST AGRIC COLLEGE	2022	NHLBI	NHLBI	$627,800